JAMA Network Open

Original Investigation | Psychiatry

Depression or Anxiety by Region, Small-Area Deprivation, and Health Care Level in Sweden

Emily White Johansson, PhD; Ahmed Nabil Shaaban, PhD; Anna Ohlis, MD; Mathias Mattsson, MSc; Lode van der Velde, MSc; Susanne Wicks, PhD; Christina Dalman, PhD; Emilie E. Agardh, PhD

Abstract

IMPORTANCE There is a lack of comprehensive evidence on the burden and management of depression or anxiety in health care systems that leverage multiple health care and prescription registers, including primary care datasets in which these conditions typically first present.

OBJECTIVE To examine the distribution of depression or anxiety among people 10 years or older in Sweden by region, small-area deprivation, and health care level of first diagnosis.

DESIGN, SETTING, AND PARTICIPANTS This nationwide, register-based, cross-sectional study included people aged 10 years or older who were living in Sweden from January 1, 2015, to December 31, 2023. Data were analyzed from January to June 2026.

EXPOSURE Small-area deprivation was defined for 5984 small geographic areas in 2015 using a published Index for Multiple Deprivation in Sweden categorized into quartiles.

MAIN OUTCOMES AND MEASURES Depression or anxiety was defined as the first diagnosis, or first prescription date if no diagnosis was recorded in the study period. Crude and age-standardized rates of depression or anxiety per 1000 persons were tabulated at national and subnational levels. Multilevel logistic regression quantified associations between small-area deprivation and the outcome adjusted for confounders. Health care level for the first diagnosis was compared by region, small-area deprivation, and patient characteristics.

RESULTS Among 8 528 198 individuals aged 10 years or older (50.2% female), 2 087 413 had recorded depression or anxiety between 2015 and 2023, corresponding to a rate of 244.8 (95% CI, 244.4-245.1) cases per 1000 persons. Higher odds of depression or anxiety were found in the most vs least deprived areas (OR, 1.18; 95% CI, 1.16-1.20). Among 1 359 196 patients with depression and anxiety between 2015 and 2017, 56.2% (95% CI, 56.1%-56.3%), 18.1% (95% CI, 18.0%-18.1%), and 3.2% (95% CI, 3.2%-3.2%) were first diagnosed in primary, outpatient, and inpatient care, respectively.

CONCLUSIONS AND RELEVANCE This cross-sectional study using administrative registers found lower rates of depression or anxiety among residents of Sweden than reported in population surveys, likely due to underestimating individuals not seeking health care. Higher odds of depression or anxiety in most vs least deprived areas could underscore the association of these conditions with socioeconomically disadvantaged contexts. Variation by region and patient characteristics in the health care level for first diagnosis merits further investigation.

JAMA Network Open. 2026;9(8):e2630530. doi:10.1001/jamanetworkopen.2026.30530

Key Points

Question What is the distribution of recorded depression or anxiety cases in Sweden by region, small-area deprivation, and health care level of the first diagnosis?

Findings In this cross-sectional study including 8 528 198 residents of Sweden, people in the most vs least deprived areas had higher odds of depression or anxiety. Among 1 359 196 patients with depression or anxiety, 56.2% received their first diagnosis in primary care with significant variation by region and patient characteristics.

Meaning These findings suggest that more consistent approaches to management of depression or anxiety are needed in Sweden, with focus on the health care needs of regions and socioeconomically deprived areas with higher burden.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Open Access. This is an open access article distributed under the terms of the CC-BY-NC-ND License, which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

Introduction

Depression and anxiety are major public health problems globally and in Sweden, affecting not only the person's health but also their social relationships, productivity, and ability to work.[1-3] In 2023, the global prevalence of depressive and anxiety disorders was estimated to reach 322 and 470 million cases respectively in that year, which represents a significant increase in the global burden since 1990.[2] These conditions disproportionately affect women and people in lower socioeconomic groups.[4-6]

In Sweden, population-based survey estimates from 2021 indicated that about 1 in 5 people reported receiving a depression diagnosis at some point in their lives, whereas 4 of 10 reported feelings of anxiety or worry.[7] Burden estimation is a well-recognized challenge in many countries due to inadequate care-seeking for mental health conditions and/or reliance on self-reports in population-based surveys.[8] Evidence suggests variable rates of treatment coverage for mental health disorders across many countries, with lower rates for common disorders such as milder depression or anxiety.[8,9] Similar patterns have been found in previous research from Sweden showing health care gaps for mental health disorders across regions where primary health care is managed, and particularly for rural communities and socioeconomically disadvantaged communities.[10-12]

In Sweden, there remains a lack of recent evidence on the distribution and management of depression and anxiety cases recorded in the health care system, including by region and small geographic areas with varying deprivation levels (herein, *small-area deprivation*). Residential areas are known to affect health outcomes, including depression or anxiety, due to contextual factors such as stress levels, crime, access to quality services, and social cohesion, as well as other contextual factors.[13,14] Prior research from Sweden has shown that depression and anxiety are more common in areas with greater socioeconomic deprivation.[10-12] However, there is no recent and comprehensive evidence of the distribution and management of these cases in the health care system leveraging multiple health care and prescription registers, including primary care datasets in which these conditions typically first present. This study aims to examine the geographical distribution and management of depression or anxiety in the health care system among persons aged 10 years or older in Sweden from 2015 to 2023, including by small-area deprivation.

Methods

Data Sources

The cross-sectional study linked national and regional registers for persons living in Sweden between 2015 and 2023 through a unique Swedish personal identification,[1] including the Total Population Register,[2] the National Prescribed Drug Register,[3] the National Patient Register providing data on the disorders and treatments from inpatient admission and specialized outpatient care,[4] and primary health care registers providing data on the disorders and treatments from primary health care centers. All registers had national coverage for the study period except primary health care registers. Primary health care data were only available from 2015 to 2017 and for 17 of 21 regions, representing 89% of the Swedish population. The study received ethical approval from the Swedish Ethical Review Authority. According to praxis for register studies, informed consent does not need to be obtained because there is no practical possibility for requesting consent from such large populations. We reported study results according to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies.

Study Population

We identified persons registered in Sweden continuously between January 1, 2015, and December 31, 2023, with a personal identification number and a primary address in 1 of 5984 small geographical divisions established by Statistics Sweden (n = 8 728 980).[15] We excluded persons younger than 10

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

years on December 31, 2023 (n = 200 782) because depression or anxiety onset typically occurs in adolescence.[3] The final study population included 8 528 198 individuals.

Depression or Anxiety

The outcome was defined as the first recorded diagnosis date of either condition in primary, specialist outpatient, or inpatient care. If no diagnosis was recorded during the study period, the first prescription date of a dispensed drug was used. eTable 1 in Supplement 1 lists the diagnosis and prescription codes to ascertain the outcome using the Swedish adaptation of the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* and the Anatomical Therapeutic Chemical classification system.

Small-Area Deprivation

In 2018, Statistics Sweden developed new demographic statistical areas (DeSO) that divided the country into approximately 6000 small geographic areas.[15] We measured small-area deprivation by applying an Index for Multiple Deprivation in Sweden (IMDIS) for 2015 to each DeSO. IMDIS has been described elsewhere.[16] Briefly, IMDIS aims to capture the deprivation level of each DeSO across 4 domains (income and capital, education, employment, and housing) using a composite measure with 15 indicators. Indicators were spatially smoothed to reduce small sample bias and enhance robustness. Domains were formed through weighted aggregation of indicators and combined into an IMDIS score using defined weights. DeSO areas were ranked from least to most deprived and categorized into quartiles at the 25th, 50th, and 75th percentiles.

Persons with depression or anxiety were assigned to a DeSO based on their primary address in the year of their first depression or anxiety diagnosis or prescription. For persons without depression or anxiety, the most frequent DeSO (mode) for the person's primary address was used if they moved during the study period. We assumed that the deprivation level for each DeSO in 2015 remained unchanged throughout the study period.[17]

Other Covariates

Region and urbanicity were defined from the DeSO code assigned to each study participant. Other covariates included sex, birthplace (Sweden, Nordic outside Sweden, 28 countries of the European Union [EU28] outside Nordic, Europe outside EU28, and other birthplace), and age at the study end date.

Statistical Analysis

Data were analyzed from January to June 2026. We calculated crude and age-standardized rates of depression or anxiety per 1000 persons aged 10 years or older. To calculate age-standardized rates, we applied the age-specific rates of depression or anxiety in each region, by urbanicity and DeSO to the age distribution of the Swedish population in 2023. Age-standardized rates of depression or anxiety for each DeSO were empirically categorized into quartile groupings and mapped alongside its assigned deprivation level.

To examine determinants of depression or anxiety, we estimated odds ratios (ORs) for the association between small-area deprivation level and the binary outcome using mixed-effects logistic regression models with covariates included as fixed effects nested within DeSO identifiers. Individual-level variables included sex, age, and birthplace, whereas contextual-level variables included small-area deprivation, region, and urbanicity. We generated the following models: model 1 (null model) contained the outcome variable and the DeSO identifier, model 2 contained individual-level variables with the DeSO identifier, and model 3 contained both individual- and contextual-level variables with the DeSO identifier. In addition, the health care level for the first diagnosis was compared for 2015 to 2017 when primary care data were available.

In the first sensitivity analysis, depression or anxiety was defined based on recorded diagnoses only during the study period. We omitted cases identified using prescribed drugs alone because

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

some antidepressants may be used to treat other conditions. In the second sensitivity analysis, depression or anxiety was ascertained without using primary health care data because these data were not available for all regions or all years in the study period. We analyzed data using Stata, version 18.1 (Stata Corporation).

Results

A total of 8 528 198 persons aged 10 years or older living in Sweden from 2015 to 2023 were included in the study (49.8% male and 50.2% female), among whom 2 087 413 (24.5%) had recorded depression or anxiety. Overall, 549 033 persons (26.6%) lived in the most deprived areas and 466 588 (21.5%) in the least deprived areas (**Table 1**).

The crude rate of depression or anxiety was 244.8 (95% CI, 244.4-245.1) cases per 1000 persons aged 10 years or older living in Sweden from 2015 to 2023 (**Figure**; eTable 2 in Supplement 1). Across regions, age-standardized rates were highest in Gotland (272.2; 95% CI, 269.4-276.0), Västra Götaland (261.3; 95% CI, 260.6-262.1), and Skåne (261.1; 95% CI, 260.3-261.9). The age-standardized rate of depression or anxiety was highest in the most deprived areas (263.9; 95% CI, 263.3-264.5) and lowest in least deprived areas (220.0; 95% CI, 219.5-220.5) (Figure).

Associations With Depression or Anxiety

In the null model, 1.6% of the variation in depression or anxiety was attributable to between-DeSO differences (variance, 0.053; SE, 0.001) (model 1 in **Table 2**), which was reduced to 1.3% after adjustment for individual-level variables (variance, 0.045; SE: 0.001) (model 2 in Table 2). In the final

Table 1. Characteristics of Study Participants With or Without Depression or Anxiety

Characteristic	Depression or anxiety		No depression or anxiety	
	No.	% (95% CI)	No.	% (95% CI)
Total	2 087 413	24.5 (24.4-2.45)	6 440 785	75.5 (75.5-75.6)
Small-area deprivation				
Very low deprivation	466 588	21.5 (21.4-21.5)	1 707 667	78.5 (78.5-78.6)
Low deprivation	524 611	24.3 (24.2-24.4)	1 634 555	75.7 (75.6-75.8)
High deprivation	547 181	25.7 (25.7-25.8)	1 579 426	74.3 (74.2-74.3)
Very high deprivation	549 033	26.6 (26.5-26.6)	1 519 137	73.5 (73.4-73.5)
Urbanicity				
Urban	1 604 624	25.0 (25.0-25.0)	4 816 070	75.0 (75.0-75.0)
Peri-urban	179 132	24.2 (24.1-24.3)	562 447	75.8 (75.7-75.9)
Rural	303 657	22.2 (22.2-22.3)	1 062 268	77.8 (77.7-77.8)
Age, y				
<20	121 138	9.2 (9.1-9.2)	1 200 625	90.8 (90.8-90.9)
20-29	291 703	27.2 (27.1-27.2)	782 320	72.8 (72.8-72.9)
30-39	371 190	30.6 (30.5-30.7)	842 068	69.4 (69.3-69.5)
40-49	343 762	29.2 (29.1-29.3)	834 653	70.8 (70.7-70.9)
50-59	337 941	26.8 (26.7-26.9)	923 418	73.2 (73.1-73.2)
≥60	621 679	25.1 (25.0-25.1)	1 857 701	74.9 (74.9-75.0)
Sex				
Male	758 076	17.8 (17.8-17.9)	3 492 675	82.2 (82.1-82.2)
Female	1 329 337	31.1 (31.0-31.1)	2 948 110	68.9 (68.9-69.0)
Birthplace				
Sweden	1 756 624	24.5 (24.4-24.5)	5 428 075	75.6 (75.5-75.6)
Nordic not Sweden	46 969	25.9 (25.7-26.1)	134 655	74.1 (73.9-74.3)
EU28 not Nordic	58 817	23.6 (23.4-23.7)	190 899	76.5 (76.3-76.6)
Europe not EU28	61 054	29.4 (29.1-29.5)	147 000	70.7 (70.5-70.9)
Other or not reported	163 949	23.3 (23.2-23.4)	540 156	76.7 (76.6-76.8)

Abbreviation: EU28, 28 countries of the European Union.

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

model, 1.7% of the variation remained unexplained after adjustment for both individual- and contextual-level variables (variance, 0.056; SE, 0.003) (model 3 in Table 2). After adjustment, compared with the least deprived areas, there were higher odds of depression or anxiety among people residing in very high (OR, 1.18; 95% CI, 1.16-1.20), high (OR, 1.15; 95% CI, 1.13-1.16), and low (OR, 1.05; 95% CI, 1.04-1.07) deprivation areas. All other individual- and contextual-level variables were also significantly associated with depression or anxiety in the final model (model 3 in Table 2).

Sensitivity Analysis

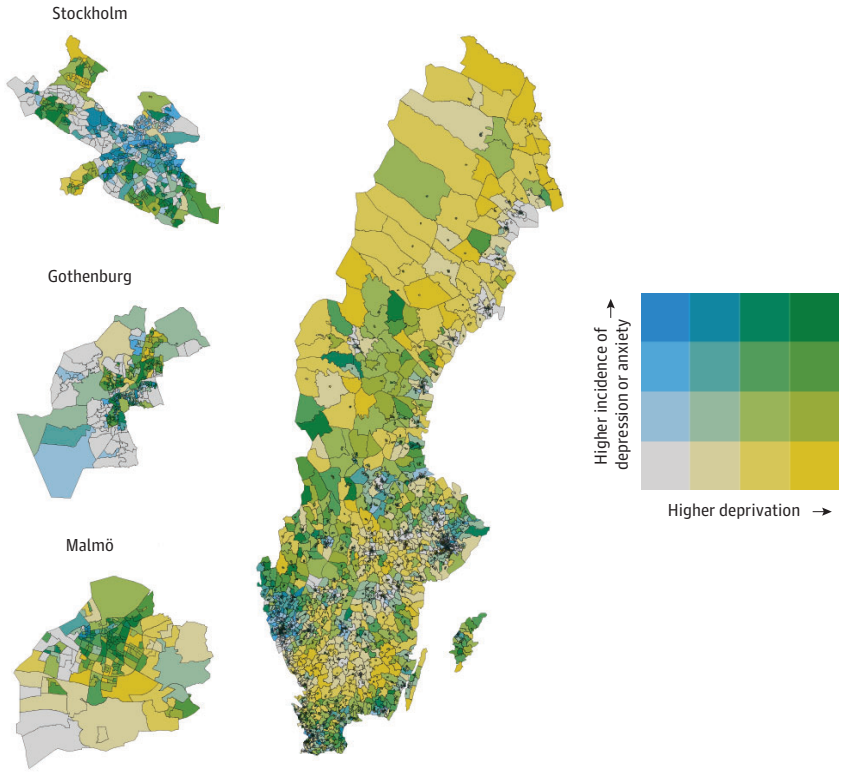
In the first sensitivity analysis, there was higher variation (3.7%) in the null model for depression or anxiety attributable to between-DeSO differences compared with the main analysis (variance, 0.127; SE, 0.002) (eTable 3 and 4 in Supplement 1), which was reduced to 1.2% in the final model (variance, 0.041; SE, 0.001). For diagnosed cases only, compared with the least deprived areas, there were higher odds of depression or anxiety among people residing in low (OR, 1.12; 95% CI, 1.10-1.13), high (OR, 1.23; 95% CI, 1.21-1.25), and very high (OR, 1.31; 95% CI, 1.29-1.33) deprivation areas.

The second sensitivity analysis ascertained the outcome without primary health care data. In the null model, there was higher variation (2.3%) in depression or anxiety attributable to between-DeSO differences compared with the main analysis (variance, 0.076; SE, 0.023) (eTable 5 in Supplement 1), which was reduced to 0.6% in the final model (variance, 0.019; SE, 0.001). In this analysis, compared with the least deprived areas, there were higher odds of depression or anxiety among people residing in low (OR, 1.08; 95% CI, 1.07-1.09), high (OR, 1.16; 95% CI, 1.15-1.17), and very high (OR, 1.21; 95% CI, 1.20-1.22) deprivation areas.

Health Care Level for the First Depression or Anxiety Diagnosis

We compared the health care level for the first recorded diagnosis of depression or anxiety between 2015 and 2017, when primary care data were available. Among the 1 359 196 patients with depression

Figure. Depression or Anxiety Rates Per 1000 Persons Aged 10 Years or Older by Small-Area Deprivation



Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

Table 2. Individual-Level and Contextual-Level Determinants of Depression or Anxiety in the Study Population and Measures of Clustering by Small Geographic Areas (DeSO) Within Sweden[a]

Characteristic	Adjusted odds ratio (95% CI)		
	Model 1	Model 2	Model 3
Contextual-level variables			
Small-area deprivation			
Very low deprivation	NA	NA	1 [Reference]
Low deprivation	NA	NA	1.05 (1.04-1.07)
High deprivation	NA	NA	1.15 (1.13-1.16)
Very high deprivation	NA	NA	1.18 (1.16-1.20)
Urbanicity			
Urban	NA	NA	0.84 (0.82-0.87)
Peri-urban	NA	NA	1.21 (1.18-1.24)
Rural	NA	NA	1 [Reference]
Region			
Stockholm	NA	NA	1 [Reference]
Uppsala	NA	NA	1.06 (1.03-1.09)
Södermanland	NA	NA	0.92 (0.89-0.96)
Östergötland	NA	NA	0.96 (0.93-0.99)
Jönkoping	NA	NA	1.03 (1.00-1.07)
Kronoberg	NA	NA	0.92 (0.88-0.96)
Kalmar	NA	NA	1.02 (0.99-1.07)
Gotland	NA	NA	1.14 (1.06-1.22)
Blekinge	NA	NA	1.10 (1.05-1.15)
Skåne	NA	NA	1.10 (1.08-1.12)
Halland	NA	NA	0.95 (0.92-0.98)
Västra götaland	NA	NA	1.14 (1.12-1.17)
Värmland	NA	NA	1.13 (1.09-1.17)
Örebro	NA	NA	1.01 (0.97-1.04)
Västmanland	NA	NA	1.06 (1.02-1.10)
Dalarna	NA	NA	1.12 (1.08-1.16)
Gävleborg	NA	NA	1.09 (1.05-1.13)
Västernorrland	NA	NA	1.00 (0.96-1.04)
Jämtland	NA	NA	0.99 (0.94-1.04)
Västerbotten	NA	NA	0.98 (0.94-1.01)
Norrbotten	NA	NA	0.99 (0.95-1.03)
Individual-level variables			
Age, y			
<20	NA	0.31 (0.31-0.31)	0.31 (0.31-0.31)
20-29	NA	1.13 (1.13-1.14)	1.14 (1.13-1.15)
30-39	NA	1.35 (1.34-1.36)	1.35 (1.35-1.36)
40-49	NA	1.31 (1.31-1.32)	1.32 (1.31-1.32)
50-59	NA	1.15 (1.14-1.15)	1.15 (1.15-1.16)
≥60	NA	1 [Reference]	1 [Reference]
Sex			
Male	NA	1 [Reference]	1 [Reference]
Female	NA	2.11 (2.11-2.12)	2.11 (2.11-2.12)
Birthplace	NA		
Sweden	NA	1 [Reference]	1 [Reference]
Nordic not Sweden	NA	0.93 (0.92-0.94)	0.93 (0.92-0.94)
EU28 not Nordic	NA	0.80 (0.80-0.81)	0.80 (0.80-0.81)
Europe not EU28	NA	0.99 (0.98-1.00)	0.98 (0.97-0.99)
Other or not reported	NA	0.75 (0.74-0.75)	0.75 (0.74-0.75)

(continued)

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

Table 2. Individual-Level and Contextual-Level Determinants of Depression or Anxiety in the Study Population and Measures of Clustering by Small Geographic Areas (DeSO) Within Sweden[a] (continued)

Characteristic	Adjusted odds ratio (95% CI)		
	Model 1	Model 2	Model 3
Model measures of clustering			
DeSO variance (SE)	0.053 (0.001)	0.045 (0.001)	0.056 (0.003)
Proportional variance change from null model, %	NA	15.1	−5.7
Intraclass correlation	0.016	0.013	0.017
Median odds ratio	1.24	1.22	1.25

Abbreviations: EU28, 28 countries of the European Union; NA, not applicable.

[a] Model 1 is the null model, model 2 includes individual-level variables, and model 3 includes both individual-level and contextual-level variables.

or anxiety during these years, 56.2% (95% CI, 56.1%-56.3%) were first diagnosed in primary care, 18.1% (95% CI, 18.0%-18.1%) in specialist outpatient care, and 3.2% (95% CI, 3.2%-3.2%) during inpatient admissions, while 22.6% (95% CI, 22.5%-22.6%) received a prescribed drug without a diagnosis recorded during the study period (**Table 3**). For patients identified based on prescribed drugs only, the diagnosis may have occurred at an earlier time with treatment for depression or anxiety continuing into the study period such that it is captured in the National Prescription Register without a concurrent diagnosis between 2015 and 2017.

In addition, there were significant variations in the proportions of patients first diagnosed with depression or anxiety in primary health care by demographic characteristics, including female (58.2%; 95% CI, 58.1%-58.3%) vs male (52.3%; 95% CI, 52.2%-52.4%), urban (57.1%; 95% CI, 57.0%-57.2%) vs rural (52.3%; 95% CI, 52.0%-52.5%), and among people living in the least deprived areas (57.4%; 95% CI, 57.3%-57.6%) vs most deprived (55.4%; 95% CI, 55.2%-55.6%). There was also significant regional variation in the distribution of first diagnoses by health care level. The regions with the highest proportions of patients first diagnosed in primary care were Stockholm (62.5%; 95% CI, 62.3%-62.7%), Jönköping (64.9%; 95% CI, 64.5%-65.3%), Västra Götaland (65.4%; 95% CI, 65.2%-65.6%), and Skåne (71.3%; 95% CI, 71.1%-71.5%) (Table 3).

Discussion

In this nationwide, register-based, cross-sectional study including 8 528 198 participants, we estimated a depression or anxiety rate of 244.8 cases per 1000 persons aged 10 years or older in Sweden between 2015 and 2023. There were higher odds of depression or anxiety for people living in the most vs least deprived areas. We found significant variation across regions and patient subgroups in the health care level where the first depression or anxiety diagnosis occurred.

The rate of depression or anxiety reported in this study was substantially lower than previous estimates for Sweden, in which approximately 1 in 4 persons reportedly received a depression diagnosis in their lifetime and 4 in 10 reported feelings of anxiety or worry in population-based survey interviews.[7] These survey interviews used broader terminology for self-reporting mental health states compared to clinical diagnoses from administrative registers. At the same time, the current research relied on administrative registers known to underestimate the burden, since it captures only people who sought care and were diagnosed or treated for the conditions.[8] In addition, our dataset only contained primary care data from 2015 to 2017 and for 17 of 21 regions, which is the health care level in which depression or anxiety cases typically first present in Sweden. Antidepressants included in the definition may be prescribed for unrelated conditions (eg, chronic pain), which could also overestimate the rate of depression or anxiety for the fraction of cases identified through prescription drugs alone. Overall, this study likely underestimated the true prevalence of depression or anxiety in the Swedish population given well-recognized health care gaps for mental health disorders, especially in rural areas.[12]

The analysis of administrative registers helped us explore health care gaps for mental health disorders, and we found substantial variation in the health care level for the first recorded depression

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

Table 3. Health Care Level for the First Diagnosis of Depression or Anxiety Between 2015 and 2017

Characteristic	Total, No.	Inpatient		Specialist outpatient		Primary[a]		Prescription only (no diagnosis)	
		No.	% (95% CI)	No.	% (95% CI)	No.	% (95% CI)	No.	% (95% CI)
Total	1 359 196	43 027	3.2 (3.2-3.2)	245 335	18.1 (18.0-18.1)	764 223	56.2 (56.1-56.3)	306 611	22.6 (22.5-22.6)
Small-area deprivation									
Very low deprivation	297 357	7560	2.5 (2.5-2.6)	48 513	16.3 (16.2-16.4)	170 795	57.4 (57.3-57.6)	70 489	23.7 (23.6-23.9)
Low deprivation	339 474	10 259	3.0 (2.9-3.1)	58 080	17.1 (17.0-17.2)	192 231	56.6 (56.4-56.8)	78 904	23.2 (23.1-23.3)
High deprivation	355 331	11 611	3.3 (3.2-3.3)	64 387	18.1 (18.0-18.2)	198 040	55.7 (55.6-55.9)	81 293	22.9 (22.7-23.0)
Very high deprivation	367 034	13 597	3.7 (3.6-3.8)	74 355	20.3 (20.1-20.4)	203 157	55.4 (55.2-55.6)	75 925	20.7 (20.6-20.8)
Age, y									
<20	20 205	436	2.2 (2.0-2.4)	10 789	53.4 (52.7-54.1)	7864	38.9 (38.3-39.6)	1116	5.5 (5.2-5.8)
20-29	165 633	5104	3.1 (3.0-3.2)	56 327	34.0 (33.8-34.2)	90 511	54.6 (54.4-54.9)	13 691	8.3 (8.1-8.4)
30-39	257 408	6793	2.6 (2.6-2.7)	52 416	20.4 (20.2-20.5)	167 875	65.2 (65.0-65.4)	30 324	11.8 (11.7-11.9)
40-49	253 970	5742	2.3 (2.2-2.3)	43 199	17.0 (16.9-17.2)	158 472	62.4 (62.2-62.6)	46 557	18.3 (18.2-18.5)
50-59	256 117	6860	2.7 (2.6-2.7)	39 596	15.5 (15.3-15.6)	149 261	58.3 (58.1-58.4)	60 400	23.6 (23.4-23.7)
≥60	405 863	18 092	4.5 (4.4-4.5)	43 008	10.6 (10.5-10.7)	190 240	46.9 (46.7-47.0)	154 523	38.1 (38.0-38.2)
Sex									
Male	461 705	16 272	3.5 (3.5-3.6)	94 623	20.5 (20.4-20.6)	241 454	52.3 (52.2-52.4)	109 356	23.7 (23.6-23.8)
Female	897 491	26 755	3.0 (2.9-3.0)	150 712	16.8 (16.7-16.9)	522 769	58.2 (58.1-58.3)	197 255	22.0 (21.9-22.1)
Birthplace									
Sweden	1 139 850	35 930	3.2 (3.1-3.2)	206 608	18.1 (18.1-18.2)	635 092	55.7 (55.6-55.8)	262 220	23.0 (22.9-23.1)
Nordic not Sweden	31 110	1390	4.5 (4.2-4.7)	4381	14.1 (13.7-14.5)	15 977	51.4 (50.8-51.9)	9362	30.1 (29.6-30.6)
EU28 not Nordic	37 834	1180	3.1 (2.9-3.3)	6008	15.9 (15.5-16.3)	22 001	58.2 (57.7-58.6)	8645	22.8 (22.4-23.3)
Europe not EU28	42 386	1123	2.6 (2.5-2.8)	7367	17.4 (17.0-17.7)	25287	59.7 (59.2-60.1)	8609	20.3 (19.9-20.7)
Other or not reported	107 967	3402	3.2 (3.0-3.3)	20 959	19.4 (19.2-19.6)	65 831	61.0 (60.7-61.2)	17 775	16.5 (16.2-16.7)
Urbanicity									
Urban	1 052 063	32 696	3.1 (3.1-3.1)	192 152	18.3 (18.2-18.3)	601 132	57.1 (57.0-57.2)	226 083	21.5 (21.4-21.6)
Peri-urban	114 862	3710	3.2 (3.1-3.3)	20 192	17.6 (17.4-17.8)	62 611	54.5 (54.2-54.8)	28 349	24.7 (24.4-24.9)
Rural	192 271	6621	3.4 (3.4-3.5)	32 991	17.2 (17.0-17.3)	100 480	52.3 (52.0-52.5)	52 179	27.1 (26.9-27.3)
Region[b]									
Stockholm	312 140	7491	2.4 (2.3-2.5)	60 199	19.3 (19.1-19.4)	195 211	62.5 (62.3-62.7)	49 239	15.8 (15.6-15.9)
Uppsala	46 104	1906	4.1 (3.9-4.3)	10 634	23.1 (22.7-23.5)	18 503	40.1 (39.7-40.6)	15 061	32.7 (32.2-33.1)
Södermanland[c]	29 978	2239	7.5 (7.2-7.8)	10 241	34.2 (33.6-34.7)	1299	4.3 (4.1-4.6)	16 199	54.0 (53.5-54.6)
Östergötland	57 817	2001	3.5 (3.3-3.6)	11 796	20.4 (20.1-20.7)	31 319	54.2 (53.8-54.6)	12 701	22.0 (21.6-22.3)
Jönkoping	48 961	1702	3.5 (3.3-3.6)	7719	15.8 (15.4-16.1)	31 763	64.9 (64.5-65.3)	7777	15.9 (15.6-15.9)
Kronoberg[c]	19 935	1102	5.5 (5.2-5.9)	5756	28.9 (28.2-29.5)	719	3.6 (3.4-3.9)	12 358	62.0 (61.3-62.7)
Kalmar	32 220	1206	3.7 (3.5-4.0)	6833	21.2 (20.8-21.7)	16850	52.3 (51.8-52.8)	7331	22.8 (22.3-23.2)
Gotland	8433	227	2.7 (2.4-3.1)	1838	21.8 (20.9-22.7)	4579	54.3 (53.2-55.4)	1789	21.2 (20.4-22.1)
Blekinge	21 428	545	2.5 (2.4-3.1)	4380	20.4 (19.9-21.0)	11391	53.2 (52.5-53.8)	5112	23.9 (23.3-24.4)
Skåne	205 403	3781	1.8 (1.8-1.9)	27 985	13.6 (13.5-13.8)	146 499	71.3 (71.1-71.5)	27 138	13.2 (13.1-13.4)
Halland[c]	32 528	2921	9.0 (8.7-9.3)	8779	27.0 (26.5-27.4)	1352	4.2 (3.9-4.4)	19 476	59.9 (59.3-60.4)
Västra götaland	254 728	7677	3.0 (2.9-.3.1)	29 792	11.7 (11.6-11.8)	166 526	65.4 (65.2-65.6)	50 733	19.9 (19.8-20.1)
Värmland	39 152	1187	3.0 (2.9-3.2)	7864	20.1 (19.7-20.5)	21143	54.0 (53.5-54.5)	8958	22.9 (22.5-23.3)
Örebro	35 096	1004	2.9 (2.7-3.0)	6935	19.8 (19.3-20.2)	17824	50.8 (50.3-51.3)	9333	26.6 (26.1-27.1)
Västmanland	31 982	1641	5.1 (5.0-5.5)	8268	25.9 (25.4-26.3)	9515	29.8 (29.3-30.2)	12 558	39.3 (38.7-39.8)
Dalarna	39 150	1083	2.8 (2.6-2.9)	7163	18.3 (18.0-18.7)	23160	59.2 (58.7-59.6)	7744	19.8 (19.4-20.2)
Gävleborg	38 714	1239	3.2 (3.2-3.3)	6493	16.8 (16.4-17.1)	22165	57.3 (56.8-57.7)	8817	22.8 (22.4-23.2)
Västernorrland	32 253	956	3.0 (2.8-3.2)	5145	16.0 (15.6-16.4)	19318	59.9 (59.4-60.4)	6834	21.2 (20.7-21.6)
Jämtland	15 455	509	3.3 (3.0-3.6)	2762	17.9 (17.3-18.5)	7333	47.4 (46.7-48.2)	4851	31.4 (30.7-32.1)
Västerbotten[c]	27 920	1470	5.3 (5.0-5.5)	10 127	36.3 (35.7-36.8)	831	3.0 (2.7-3.2)	15 492	55.5 (54.9-56.1)
Norrbotten	29 799	1140	3.8 (3.6-4.0)	4626	15.5 (15.1-15.9)	16 923	56.8 (56.2-57.4)	7110	23.9 (23.4-24.3)

Abbreviation: EU28, 28 countries of the European Union.

[a] Primary health care data were only available between 2015 and 2017 and for only 17 of 21 regions that represent 89% of the Swedish population.

[b] Regional values refer to the participant's region of primary residence, not facility location.

[c] Primary health care data were not available for these regions. The totals presented refer to the number of individuals with primary residence in the region receiving a diagnosis at a primary health care visit that took place outside that region.

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

or anxiety diagnosis by region and patient characteristics. Among the 17 of 21 regions with primary care data for this subanalysis, Stockholm, Jönköping, Västra Götaland, and Skåne had more than 60% of cases first diagnosed in primary care compared with lower proportions in other regions. In these regions, most cases would be expected to be diagnosed in primary care given the local organization of mental health service delivery and long-standing digital registration of primary health care data. Further research should examine whether observed regional variations were due to data quality of primary care registers or reflect how mental health care is delivered. We also found a higher proportion of urban residents, women, and people living in least deprived areas first diagnosed in primary care. These groups may have better access to and increased contacts with primary care in their local areas, particularly for mental health services.[12] Prior research from Sweden has shown variations in the uptake of mental health services for these population groups.[4,12,18,19]

Multilevel analysis showed that depression or anxiety was positively associated with living in the most vs least deprived areas, and the odds ratios increased as deprivation level increased. This finding is consistent with prior research from Sweden and other settings[20-24] and could reflect increased susceptibility to depression and anxiety in areas characterized by greater socioeconomic disadvantage.[14,22,24] At the same time, there could be reverse causation such that people with mental health disorders more often reside in more deprived areas due to lower income or lack of resources to live elsewhere. This positive association was found in both sensitivity analyses, which further supports main study findings.

In multilevel analyses, we found limited variation in depression or anxiety attributable to between-DeSO variation.[25] The limited between-DeSO variation in our results supports the recognized importance of individual factors (eg, age, sex, and birthplace) in depression or anxiety.[3,5,12] There may also be limited clustering at DeSO levels because these small geographic areas may not overlap with residential areas or reflect administrative levels for health care delivery. Our analysis showed that region, urbanicity, and small-area deprivation were important contextual determinants of depression or anxiety consistent with previous research, underscoring geographic disparities and socioeconomic inequalities in the use of mental health services within Sweden.[4,12,18,19]

Sensitivity analyses indicated low variation in depression or anxiety attributable to between-DeSO variation similar to main findings. In the sensitivity analyses, a lower proportion of the variation remained unexplained in the final models compared with main results, which could indicate more consistent approaches to care across areas when the outcome is not ascertained from prescribed medication alone or diagnosed through local primary clinics. At the same time, in both sensitivity analyses, there were higher odds of depression or anxiety among people living in the most vs least deprived areas, similar to the main analysis. The magnitude of effect was higher in both sensitivity analyses, which again could suggest better access to or use of prescribed medications or local primary care for depression or anxiety symptoms in persons living in better-resourced areas within Sweden.[4,12,18,19]

Limitations

This study has several limitations; therefore, findings should be interpreted given study limitations. First, depression or anxiety prevalence in this study underestimates the true population burden due to inadequate care-seeking and limitations in service delivery for mental health disorders. Age at first depression or anxiety diagnosis is typically at older than 10 years, which may further underestimate prevalence estimates. Second, regional primary care registers were available for only 17 of 21 regions between 2015 and 2017, which further contributes to underestimation. At the same time, nearly one-quarter of cases were ascertained from prescription registers alone, and some antidepressants could be used to treat other conditions, leading to overestimation for this fraction of cases. Third, contextual factors, such as small-area deprivation, have measurement challenges, including defining the area and using composite indexes that obscure differences in areas even within the same deprivation level. Fourth, we did not account for potential spatial autocorrelation of the outcome that could inflate the precision of estimated coefficients if spatial dependence is present, although

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

we adjusted for region and urbanicity in multilevel analyses to partially account for this issue. Fifth, small-area deprivation for each DeSO was measured in 2015, and the quartile level was assumed to remain unchanged throughout the study period.

Conclusions

In this nationwide, register-based, cross-sectional study, we found lower rates of recorded depression or anxiety compared to population survey estimates. Higher odds of depression or anxiety in the most vs least deprived areas could underscore the relationship between mental health conditions and socioeconomically disadvantaged areas. Substantial variation by region and patient characteristics in the health care level for first diagnosis merits further investigation to better understand reasons for differences within Sweden.

ARTICLE INFORMATION

Accepted for Publication: July 2, 2026.

Published: August 24, 2026. doi:10.1001/jamanetworkopen.2026.30530

Open Access: This is an open access article distributed under the terms of the CC-BY-NC-ND License, which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies. © 2026 Johansson EW et al. *JAMA Network Open*.

Corresponding Author: Emily White Johansson, PhD, Department of Women's and Children's Health, Global Health and Migration Unit, Uppsala University, Akademiska sjukhuset, 751 85 Uppsala, Sweden (emily.johansson@uu.se).

Author Affiliations: Department of Global Public Health, Karolinska Institutet, Stockholm, Sweden (Johansson, Shaaban, Ohlis, Mattsson, van der Velde, Wicks, Dalman, Agardh); Global Health and Migration Unit, Department of Women's and Children's Health, Uppsala University, Uppsala, Sweden (Johansson); Centre for Epidemiology and Community Medicine, Region Stockholm, Stockholm, Sweden (Ohlis, Wicks, Dalman).

Author Contributions: Dr Johansson and Mr Mattsson had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Johansson and Shaaban contributed equally to this work.

Concept and design: Johansson, Shaaban, Mattsson, Wicks, Dalman, Agardh.

Acquisition, analysis, or interpretation of data: Johansson, Shaaban, Ohlis, Mattsson, van der Velde, Wicks, Agardh.

Drafting of the manuscript: Johansson, Shaaban, Dalman.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Johansson, Shaaban, Mattsson.

Obtained funding: Agardh.

Administrative, technical, or material support: Shaaban, van der Velde.

Supervision: Wicks, Dalman, Agardh.

Conflict of Interest Disclosures: Dr Mattsson reported receiving grants from Försäkringskassan and FORTE during the conduct of the study. Dr Agardh reported receiving grants from Försäkringskassan and the Swedish Research Council for Health, Working Life and Welfare during the conduct of the study. No other disclosures were reported.

Funding/Support: This work was supported by Försäkringskassan and the Swedish Research Council for Health, Working Life and Welfare.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

REFERENCES

1. Sobocki P, Jönsson B, Angst J, Rehnberg C. Cost of depression in Europe. *J Ment Health Policy Econ*. 2006;9(2):87-98.

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

2. GBD Disease and Injury and Risk Factor Collaborators. Burden for 375 diseases and injuries, risk-attributable burden for 88 risk factors in 204 countries and territories, including 660 subnational locations, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet*. 2025;406(10513):1873-1922. doi:10.1016/S0140-6736(25)01637-X

3. Cassano P, Fava M. Depression and public health: an overview. *J Psychosom Res*. 2002;53(4):849-857. doi:10.1016/S0022-3999(02)00304-5

4. Kosidou K, Dalman C, Lundberg M, Hallqvist J, Isacsson G, Magnusson C. Socioeconomic status and risk of psychological distress and depression in the Stockholm Public Health Cohort: a population-based study. *J Affect Disord*. 2011;134(1-3):160-167. doi:10.1016/j.jad.2011.05.024

5. Remes O, Wainwright N, Surtees P, Lafortune L, Khaw KT, Brayne C. Sex differences in the association between area deprivation and generalised anxiety disorder: British population study. *BMJ Open*. 2017;7(5):e013590. doi:10.1136/bmjopen-2016-013590

6. Folkhälsomyndigheten. *Towards and Good and Equitable Health—A Framework for Implementing and Monitoring the National Public Health Policy*. Folkhälsomyndigheten; 2021.

7. Folkhälsomyndigheten. *Hur mår vi i Sverige?* Folkhälsomyndigheten; 2026.

8. Barbui C, Alonso J, Chisholm D, et al. Mental health service coverage and gaps among adults in Europe: a systematic review. *Lancet Reg Health Eur*. 2025;57:101458. doi:10.1016/j.lanepe.2025.101458

9. Moitra M, Santomauro D, Collins PY, et al. The global gap in treatment coverage for major depressive disorder in 84 countries from 2000-2019: a systematic review and Bayesian meta-regression analysis. *PLoS Med*. 2022;19(2):e1003901. doi:10.1371/journal.pmed.1003901

10. Wahlbeck K, Westman J, Nordentoft M, Gissler M, Laursen TM. Outcomes of Nordic mental health systems: life expectancy of patients with mental disorders. *Br J Psychiatry*. 2011;199(6):453-458. doi:10.1192/bjp.bp.110.085100

11. Meyer J, Engström S, Löthberg M, et al. Sociodemographic predictors of mental health service utilization among young adults with support in daily living in Sweden: a register-based study. *BMC Psychiatry*. 2025;25(1):580. doi:10.1186/s12888-025-07046-1

12. Olsson S, Hensing G, Burström B, Löve J. Unmet need for mental healthcare in a population sample in Sweden: a cross-sectional study of inequalities based on gender, education, and country of birth. *Community Ment Health J*. 2021;57(3):470-481. doi:10.1007/s10597-020-00668-7

13. Stafford M, Marmot M. Neighbourhood deprivation and health: does it affect us all equally? *Int J Epidemiol*. 2003;32(3):357-366. doi:10.1093/ije/dyg084

14. Marmot M, Bell R. Fair society, healthy lives. *Public Health*. 2012;126(suppl 1):S4-S10. doi:10.1016/j.puhe.2012.05.014

15. Statistics Sweden. *DeSO i Statistikdatabasen*. Statistics Sweden; 2025.

16. van der Velde L, Shabaan AN, Mattsson M, et al. An index of multiple deprivation in Sweden: measuring area-level socio-economic inequalities. *Eur J Public Health*. 2025;35(6):1107-1114. doi:10.1093/eurpub/ckaf138

17. Vogiazides L, Mondani H. Neighbourhood trajectories in Stockholm: investigating the role of mobility and in situ change. *Appl Geogr*. 2023;150:102823. doi:10.1016/j.apgeog.2022.102823

18. Brydsten A, Hammarström A, San Sebastian M. Health inequalities between employed and unemployed in northern Sweden: a decomposition analysis of social determinants for mental health. *Int J Equity Health*. 2018;17(1):59. doi:10.1186/s12939-018-0773-5

19. Goicolea I, Hultstrand Ahlin C, Waenerlund AK, et al. Accessibility and factors associated with utilization of mental health services in youth health centers: a qualitative comparative analysis in northern Sweden. *Int J Ment Health Syst*. 2018;12:69. doi:10.1186/s13033-018-0249-4

20. Crump C, Sundquist K, Sundquist J, Winkleby MA. Neighborhood deprivation and psychiatric medication prescription: a Swedish national multilevel study. *Ann Epidemiol*. 2011;21(4):231-237. doi:10.1016/j.annepidem.2011.01.005

21. Chaix B, Leyland AH, Sabel CE, et al. Spatial clustering of mental disorders and associated characteristics of the neighbourhood context in Malmö, Sweden, in 2001. *J Epidemiol Community Health*. 2006;60(5):427-435. doi:10.1136/jech.2005.040360

22. Lofors J, Sundquist K. Low-linking social capital as a predictor of mental disorders: a cohort study of 4.5 million Swedes. *Soc Sci Med*. 2007;64(1):21-34. doi:10.1016/j.socscimed.2006.08.024

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026

23. Richardson R, Westley T, Gariépy G, Austin N, Nandi A. Neighborhood socioeconomic conditions and depression: a systematic review and meta-analysis. *Soc Psychiatry Psychiatr Epidemiol*. 2015;50(11):1641-1656. doi:10.1007/s00127-015-1092-4

24. Lofors J, Ramírez-León V, Sundquist K. Neighbourhood income and anxiety: a study based on random samples of the Swedish population. *Eur J Public Health*. 2006;16(6):633-639. doi:10.1093/eurpub/ckl026

25. Merlo J, Chaix B, Ohlsson H, et al. A brief conceptual tutorial of multilevel analysis in social epidemiology: using measures of clustering in multilevel logistic regression to investigate contextual phenomena. *J Epidemiol Community Health*. 2006;60(4):290-297. doi:10.1136/jech.2004.029454

SUPPLEMENT 1.

eTable 1. ICD-10-SE and ATC Codes for Depression or Anxiety Measurement
eTable 2. Depression or Anxiety Rates Per 1000 Persons Aged Ten Years or Older in 2015-2023
eTable 3. Characteristics of Study Participants With or Without Diagnosed Depression or Anxiety
eTable 4. Associations Between Small-Area Deprivation and Depression or Anxiety Ascertained Through Diagnoses Only
eTable 5. Associations Between Small-Area Deprivation and Depression or Anxiety Ascertained Without Using Primary Care Data

SUPPLEMENT 2.

Data Sharing Statement

Downloaded from jamanetwork.com by Uppsala University user on 09/16/2026